Match! Phase Analysis Report

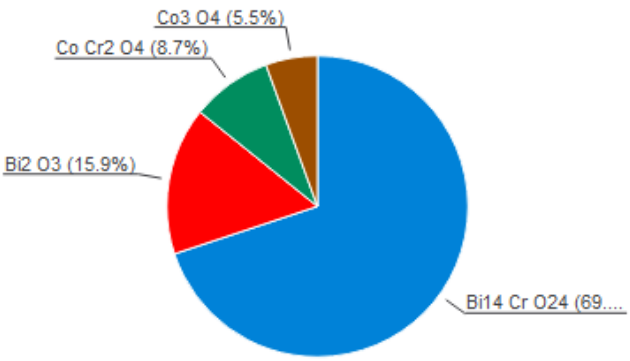
Sample: Commander Sample ID

Sample Data

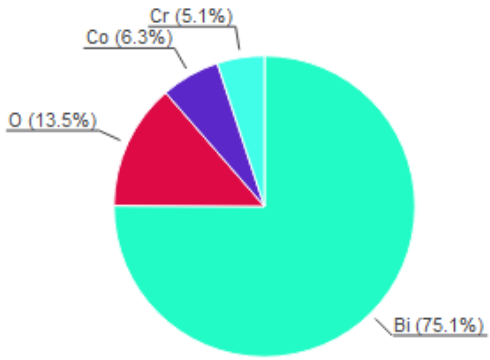
File name	Bi2CoCrO6.raw
File path	C:/Users/Daniel León/OneDrive/Documentos/AAMaster/Física Experimental Avanzada/Proyecto/DRX
Data collected	10/14/2026, 13:46:05
Data range	10.299° - 75.273°
Original data range	10.000° - 75.002°
Number of points	2183
Step size	0.030
Rietveld refinement converged	Yes
Alpha2 subtracted	No
Background subtr.	No
Data smoothed	No
2theta correction	0.15702°
Specimen displacement correction (Bragg-Brentano geometry)	T = (-s/R) = -0.0012403
Radiation	X-rays
Wavelength	1.541874 Å

Analysis Results

Phase composition (Weight %) calc. by autom. Rietveld ...



Elemental composition (Weight %) calc. by autom. Riet...



Index	*Amount (%)*	*Name*	*Formula sum*
A	69.9		Bi14 Cr O24
B	15.9		Bi2 O3
C	8.7	CoCr2O4	Co Cr2 O4
D	5.5		Co3 O4
	3.7	*Unidentified peak area*	

Element	*Amount (weight %)*
Bi	75.1%
O	*13.5%(*)*
Co	6.3%
Cr	5.1%
**LE (sum)*	*13.5%*

Amounts calculated by automatic Rietveld refinement

Details of identified phases

A: Bi14 Cr O24 (69.9 %)*

Formula sum	Bi14 Cr O24
Entry number	96-152-2145
Figure-of-Merit (FoM)	0.726354*
Total number of peaks	330
Peaks in range	233
Peaks matched	93
Intensity scale factor	0.93
2theta correction	-0.006°
Space group	I 4/m
Crystal system	tetragonal
Unit cell	a= 8.6840 Å c= 17.2380 Å
I/Ic	22.49
Calc. density	8.590 g/cm³
Refinement method	Rietveld
Reference	Warda S.A., Kesper U., Pietzuch W., Massa W., Reinen D., "Color and constitution of Cr(VI)-doped Bi2 O3 phases: the structure ofBi14 Cr O24", Journal of Solid State Chemistry **149**, 209-217 (2000)

B: Bi2 O3 (15.9 %)*

Formula sum	Bi2 O3
Entry number	96-154-0060
Figure-of-Merit (FoM)	0.483210*
Total number of peaks	164
Peaks in range	60
Peaks matched	21
Intensity scale factor	0.13
2theta correction	-0.041°
Space group	I 2 3
Crystal system	cubic
Unit cell	a= 10.2680 Å
I/Ic	13.89
Calc. density	9.291 g/cm³
Refinement method	Rietveld
Reference	Harwig H.A., "On the structure of bismuthsesquioxide: the alpha, beta, gamma and deltaphase", Zeitschrift fuer Anorganische und Allgemeine Chemie **444**, 151-166 (1978)

C: CoCr2O4 (8.7 %)*

Formula sum	Co Cr2 O4
Entry number	96-591-0125
Figure-of-Merit (FoM)	0.665659*
Total number of peaks	70
Peaks in range	24
Peaks matched	15
Intensity scale factor	0.06
2theta correction	-0.102°
Space group	F d -3 m
Crystal system	cubic
Unit cell	a= 8.3100 Å
I/Ic	11.94
Calc. density	5.413 g/cm³
Refinement method	Rietveld
Reference	Wyckoff R. W. G., "Page 290 from the second edition of Structure of Crystals by WyckoffR W G. Published by The Chemical Catalog Company, INC, New York in 1931& http://jcrystal.com/steffenweber/gallery/StructureTypes/st4.html", The second edition of Structure of Crystals, 290-290 (1931)

D: Co3 O4 (5.5 %)*

Formula sum	Co3 O4
Entry number	96-591-0032
Figure-of-Merit (FoM)	0.526870*
Total number of peaks	68
Peaks in range	24
Peaks matched	14
Intensity scale factor	0.04
2theta correction	0.104°
Space group	F d -3 m
Crystal system	cubic
Unit cell	a= 8.1690 Å
I/Ic	12.54 (Source: Unknown)
Calc. density	5.868 g/cm³
Refinement method	Rietveld
Reference	The Crystal Structure of Solids, 129-129

(*)2theta values have been shifted internally for the calculation of the amounts, the intensity scaling factors as well as the figure-of-merit (FoM), due to the active search-match option 'Automatic zero point adaption'.

Search-Match

Settings

Reference database used	COD-Inorg 2025.12.01
Method	Profile fitting search-match (PFSM)
Automatic zeropoint adaptation	Yes
Downgrade entries with low scaling factors	Yes
Minimum figure-of-merit (FoM)	0.60
2theta window for peak corr.	0.30 deg.
Minimum rel. int. for peak corr.	0
Parameter/influence 2theta	0.50
Parameter/influence intensities	0.50
Parameter multiple/single phase(s)	0.50

Selection Criteria

Elements:

Elements that may be present or not:	O, Cr, Co, Bi
Elements that must NOT be present:	All elements not mentioned above

Peak List

No.	2theta [°]	d [Å]	I/I0 (peak height)	Counts (peak area)	FWHM	Matched
1	11.28	7.8453	17.54	60.63	0.3370	
2	11.31	7.8262	8.77	30.31	0.3370	
3	12.61	7.0193	14.09	35.86	0.2481	B
4	12.64	7.0022	7.05	17.93	0.2481	
5	14.27	6.2052	43.32	79.40	0.1787	
6	14.31	6.1901	21.66	39.70	0.1787	
7	15.50	5.7159	3.69	48.22	1.2736	
8	18.35	4.8357	15.88	16.63	0.1021	
9	18.39	4.8239	7.94	8.31	0.1021	
10	18.71	4.7419	12.06	38.29	0.3096	A,C
11	18.76	4.7303	6.03	19.14	0.3096	A,D
12	27.83	3.2056	457.70	838.94	0.1787	B
13	27.89	3.1994	1000.00	1832.94	0.1787	
14	27.90	3.1977	228.85	419.47	0.1787	A
15	27.96	3.1916	500.00	916.47	0.1787	A
16	30.56	2.9250	30.79	104.07	0.3296	B
17	30.64	2.9178	15.39	52.03	0.3296	C
18	31.02	2.8832	146.88	345.19	0.2292	
19	31.10	2.8761	73.44	172.59	0.2292	A,D
20	31.55	2.8357	12.87	68.71	0.5207	A
21	31.63	2.8288	6.43	34.36	0.5207	A
22	32.05	2.7929	6.11	11.20	0.1787	
23	32.72	2.7369	303.99	557.19	0.1787	
24	32.80	2.7302	151.99	278.60	0.1787	A,B
25	36.03	2.4926	75.72	232.59	0.2995	C
26	36.13	2.4864	37.86	116.29	0.2995	A,C
27	36.55	2.4586	17.17	67.81	0.3852	
28	36.64	2.4526	8.58	33.90	0.3852	D
29	37.04	2.4272	24.97	86.91	0.3394	
30	37.13	2.4212	12.49	43.45	0.3394	B
31	43.76	2.0686	14.98	36.01	0.2345	A,C
32	43.88	2.0635	7.49	18.01	0.2345	C
33	45.49	1.9941	98.80	362.20	0.3575	B
34	45.60	1.9892	49.40	181.10	0.3575	
35	45.75	1.9832	56.22	240.44	0.4171	A
36	45.87	1.9783	28.11	120.22	0.4171	A
37	46.88	1.9381	48.67	219.84	0.4405	
38	47.00	1.9333	24.33	109.92	0.4405	A
39	53.01	1.7275	47.63	281.72	0.5768	A
40	53.15	1.7233	23.81	140.86	0.5768	A
41	54.18	1.6928	11.72	35.96	0.2992	A,C
42	54.33	1.6887	5.86	17.98	0.2992	A,C
43	55.22	1.6634	92.31	447.63	0.4729	A,B,D
44	55.37	1.6593	46.15	223.82	0.4729	A,B,D
45	56.79	1.6212	4.90	8.98	0.1787	B
46	57.20	1.6105	44.33	161.43	0.3551	B
47	57.36	1.6065	22.17	80.72	0.3551	A
48	57.77	1.5960	25.20	87.05	0.3368	A,C
49	57.92	1.5921	12.60	43.53	0.3368	A,C
50	63.36	1.4680	19.67	77.91	0.3862	C
51	63.54	1.4643	9.84	38.95	0.3862	A,C
52	63.88	1.4572	13.02	55.19	0.4133	A
53	64.17	1.4513	4.69	19.88	0.4133	A,B
54	64.38	1.4471	8.35	31.33	0.3658	A,B
55	64.56	1.4435	4.18	15.67	0.3658	A,D
56	64.87	1.4374	11.44	20.98	0.1787	A,D
57	65.05	1.4339	5.72	10.49	0.1787	A
58	68.22	1.3748	12.67	65.02	0.5003	A,C,D
59	68.41	1.3714	6.34	32.51	0.5003	A,B
60	73.33	1.2910	22.84	161.35	0.6889	A,D
61	73.55	1.2878	11.42	80.67	0.6889	A,D
62	74.11	1.2793	8.81	16.15	0.1787	A,B
63	74.33	1.2762	4.41	8.08	0.1787	

Refinement using FullProf

Method	Rietveld refinement
Automatic refinement	Yes
Final weighted average Bragg R-factor	18.9
Final reduced chi^2	9.6
FullProf comment	Your Chi-square is NOT HIGH but the model could be improved => Your Chi-square is NOT HIGH but the model could be improved if the statistic is good enough!

Parameter List for Refinement

Parameter	Final value	Refined
Bck_0_pat1	1.00872	+
Scale_ph1_pat1	5.20488E-06	+
Bover_ph1_pat1	2.41638	+

U-Cagl_ph1_pat1	0.0522277	+
Cell_A_ph1_pat1	8.7178	+
Cell_C_ph1_pat1	17.4414	+
Scale_ph2_pat1	5.33284E-06	+
Bover_ph2_pat1	-4.54603	+
U-Cagl_ph2_pat1	0.0697236	+
Cell_A_ph2_pat1	8.32608	+
Scale_ph3_pat1	3.49878E-06	+
Bover_ph3_pat1	-5.22893	+
U-Cagl_ph3_pat1	0.0782083	+
Cell_A_ph3_pat1	8.17482	+
Scale_ph4_pat1	1.57743E-06	+
Bover_ph4_pat1	13.0016	+
U-Cagl_ph4_pat1	-0.108164	+
Cell_A_ph4_pat1	10.3307	+
Zero_pat1	-0.157021	
SyCos_pat1	-0.14213	
SySin_pat1	0	
Lambda_pat1	1.54187	
P0_mabs_pat1	0	
Cp_mabs_pat1	0	
Tau_mabs_pat1	0.1	
Bck_1_pat1	0	
Bck_2_pat1	0	
Bck_3_pat1	0	
Strain1_ph1_pat1	0	
Strain2_ph1_pat1	0	
Strain3_ph1_pat1	0	
G-Size_ph1_pat1	0	
L-Size_ph1_pat1	0	
Y-cos_ph1_pat1	0	
EtaRght0_ph1_pat1	0	
X-tan_ph1_pat1	0	
V-Cagl_ph1_pat1	0	
W-Cagl_ph1_pat1	0.031948	
EtaPV_ph1_pat1	1.25364	
Cell_B_ph1_pat1	8.71801	
Cell_D_ph1_pat1	90	
Cell_E_ph1_pat1	90	
Cell_F_ph1_pat1	90	
Or1_ph1_pat1	0	
Or2_ph1_pat1	0	
Asym1_ph1_pat1	0	
Asym2_ph1_pat1	0	
Asym3_ph1_pat1	0	
Asym4_ph1_pat1	0	
X_O1_ph1	0.1225	
Y_O1_ph1	0.1828	
Z_O1_ph1	0.4218	
Biso_O1_ph1	1	
Occ_O1_ph1	1	
X_Bi2_ph1	0.7866	
Y_Bi2_ph1	0.5624	
Z_Bi2_ph1	0	
Biso_Bi2_ph1	1	
Occ_Bi2_ph1	1	
X_O3_ph1	0.199	
Y_O3_ph1	0.017	
Z_O3_ph1	0	
Biso_O3_ph1	1	
Occ_O3_ph1	0.23	
X_O4_ph1	0	
Y_O4_ph1	0.5	
Z_O4_ph1	0.3775	
Biso_O4_ph1	1	
Occ_O4_ph1	1	
X_O5_ph1	0.7476	
Y_O5_ph1	0.0772	
Z_O5_ph1	0.2562	
Biso_O5_ph1	1	
Occ_O5_ph1	1	
X_Cr6_ph1	0	
Y_Cr6_ph1	0	
Z_Cr6_ph1	0.0188	
Biso_Cr6_ph1	1	
Occ_Cr6_ph1	0.5	
X_Bi7_ph1	0	
Y_Bi7_ph1	0	
Z_Bi7_ph1	0.6577	
Biso_Bi7_ph1	1	
Occ_Bi7_ph1	1	
X_O8_ph1	0	
Y_O8_ph1	0	
Z_O8_ph1	0.111	

Biso_O8_ph1	1
Occ_O8_ph1	0.5
X_O9_ph1	0.153
Y_O9_ph1	0.104
Z_O9_ph1	0
Biso_O9_ph1	1
Occ_O9_ph1	0.15
X_Bi10_ph1	0.7981
Y_Bi10_ph1	0.6056
Z_Bi10_ph1	0.6691
Biso_Bi10_ph1	1
Occ_Bi10_ph1	1
X_O11_ph1	0.154
Y_O11_ph1	-0.065
Z_O11_ph1	0.028
Biso_O11_ph1	1
Occ_O11_ph1	0.19
Strain1_ph2_pat1	0
Strain2_ph2_pat1	0
Strain3_ph2_pat1	0
G-Size_ph2_pat1	0
L-Size_ph2_pat1	0
Y-cos_ph2_pat1	0
EtaRght0_ph2_pat1	0
X-tan_ph2_pat1	0
V-Cagl_ph2_pat1	0
W-Cagl_ph2_pat1	0.031948
EtaPV_ph2_pat1	1.64609
Cell_B_ph2_pat1	8.32709
Cell_C_ph2_pat1	8.32709
Cell_D_ph2_pat1	90
Cell_E_ph2_pat1	90
Cell_F_ph2_pat1	90
Or1_ph2_pat1	0
Or2_ph2_pat1	0
Asym1_ph2_pat1	0
Asym2_ph2_pat1	0
Asym3_ph2_pat1	0
Asym4_ph2_pat1	0
X_Co1_ph2	0
Y_Co1_ph2	0
Z_Co1_ph2	0
Biso_Co1_ph2	1
Occ_Co1_ph2	1
X_Cr2_ph2	0.375
Y_Cr2_ph2	0.375
Z_Cr2_ph2	0.375
Biso_Cr2_ph2	1
Occ_Cr2_ph2	1
X_O3_ph2	0.387
Y_O3_ph2	0.387
Z_O3_ph2	0.387
Biso_O3_ph2	1
Occ_O3_ph2	1
Strain1_ph3_pat1	0
Strain2_ph3_pat1	0
Strain3_ph3_pat1	0
G-Size_ph3_pat1	0
L-Size_ph3_pat1	0
Y-cos_ph3_pat1	0
EtaRght0_ph3_pat1	0
X-tan_ph3_pat1	0
V-Cagl_ph3_pat1	0
W-Cagl_ph3_pat1	0.031948
EtaPV_ph3_pat1	2.63933
Cell_B_ph3_pat1	8.17315
Cell_C_ph3_pat1	8.17315
Cell_D_ph3_pat1	90
Cell_E_ph3_pat1	90
Cell_F_ph3_pat1	90
Or1_ph3_pat1	0
Or2_ph3_pat1	0
Asym1_ph3_pat1	0
Asym2_ph3_pat1	0
Asym3_ph3_pat1	0
Asym4_ph3_pat1	0
X_Co1_ph3	0
Y_Co1_ph3	0
Z_Co1_ph3	0
Biso_Co1_ph3	1
Occ_Co1_ph3	1
X_Co2_ph3	0.375
Y_Co2_ph3	0.375
Z_Co2_ph3	0.375

Biso_Co2_ph3	1
Occ_Co2_ph3	1
X_O3_ph3	0.387
Y_O3_ph3	0.387
Z_O3_ph3	0.387
Biso_O3_ph3	1
Occ_O3_ph3	1
Strain1_ph4_pat1	0
Strain2_ph4_pat1	0
Strain3_ph4_pat1	0
G-Size_ph4_pat1	0
L-Size_ph4_pat1	0
Y-cos_ph4_pat1	0
EtaRght0_ph4_pat1	0
X-tan_ph4_pat1	0
V-Cagl_ph4_pat1	0
W-Cagl_ph4_pat1	0.031948
EtaPV_ph4_pat1	-0.835113
Cell_B_ph4_pat1	10.3316
Cell_C_ph4_pat1	10.3316
Cell_D_ph4_pat1	90
Cell_E_ph4_pat1	90
Cell_F_ph4_pat1	90
Or1_ph4_pat1	0
Or2_ph4_pat1	0
Asym1_ph4_pat1	0
Asym2_ph4_pat1	0
Asym3_ph4_pat1	0
Asym4_ph4_pat1	0
X_Bi1_ph4	0.858
Y_Bi1_ph4	0.687
Z_Bi1_ph4	0.972
Biso_Bi1_ph4	1
Occ_Bi1_ph4	1
X_O2_ph4	0.148
Y_O2_ph4	0.148
Z_O2_ph4	0.148
Biso_O2_ph4	1
Occ_O2_ph4	0.975
X_O3_ph4	0.716
Y_O3_ph4	0.716
Z_O3_ph4	0.716
Biso_O3_ph4	1
Occ_O3_ph4	0.975
X_Bi4_ph4	0
Y_Bi4_ph4	0
Z_Bi4_ph4	0
Biso_Bi4_ph4	1
Occ_Bi4_ph4	1
X_O5_ph4	0.921
Y_O5_ph4	0.738
Z_O5_ph4	0.514
Biso_O5_ph4	1
Occ_O5_ph4	0.975

Integrated Profile Areas

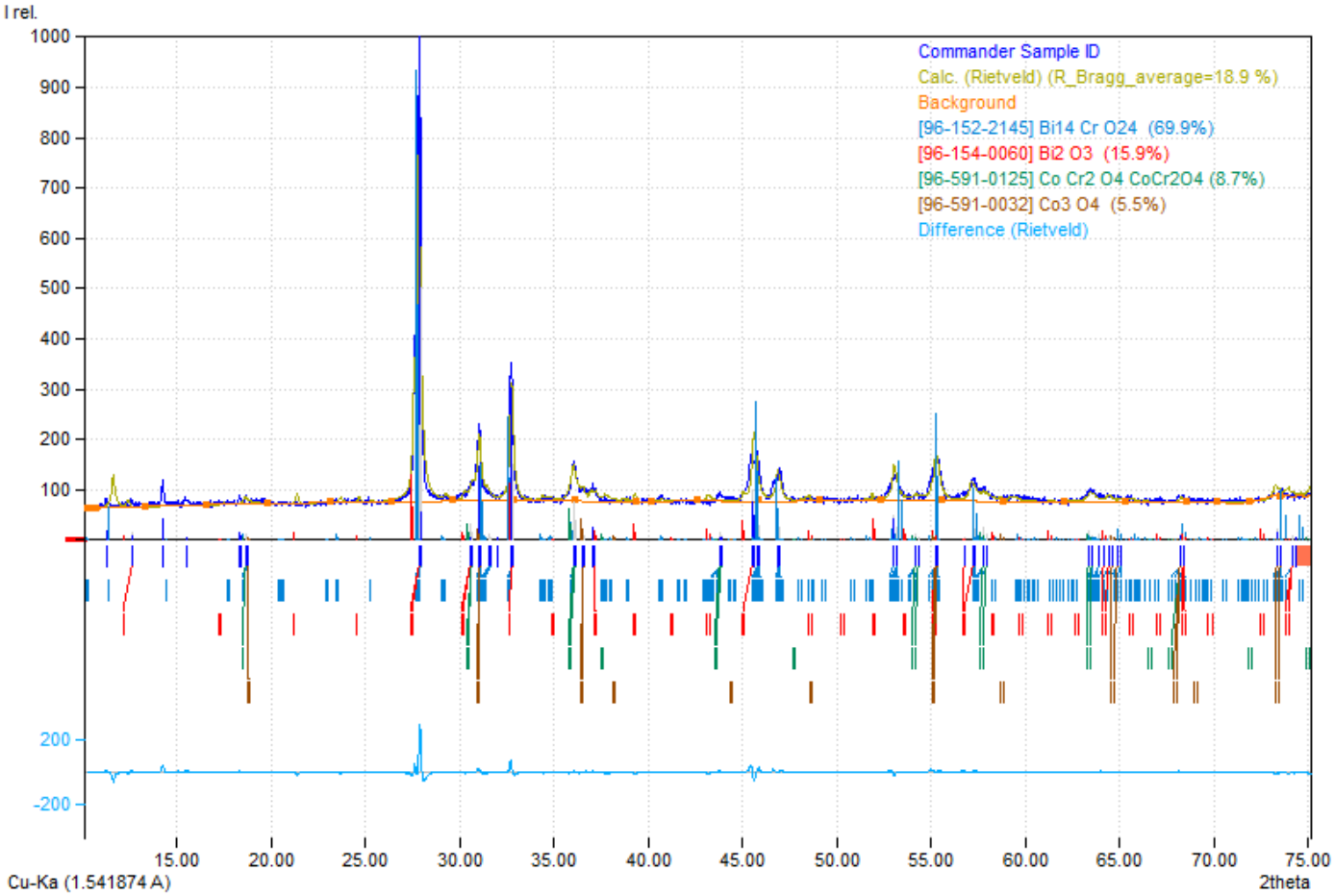
Based on Rietveld refinement

Profile area	***Counts***	***Amount***
Overall diffraction profile	2201663	100.00%
Background radiation	1857295	84.36%
Diffraction peaks	344368	15.64%
Peak area belonging to selected phases	263305	11.96%
Peak area of phase A (Bi14 Cr O24)	*104308*	*4.74%*
Peak area of phase B (Bi2 O3)	*53446*	*2.43%*
Peak area of phase C (CoCr2O4)	*53127*	*2.41%*
Peak area of phase D (Co3 O4)	*52425*	*2.38%*
Unidentified peak area	81063	3.68%

Peak Residuals

Peak data	***Counts***	***Amount***
Overall peak intensity	10059	100.00%
Peak intensity belonging to selected phases	8723	86.72%
Unidentified peak intensity	1335	13.28%

Diffraction Pattern Graphics



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